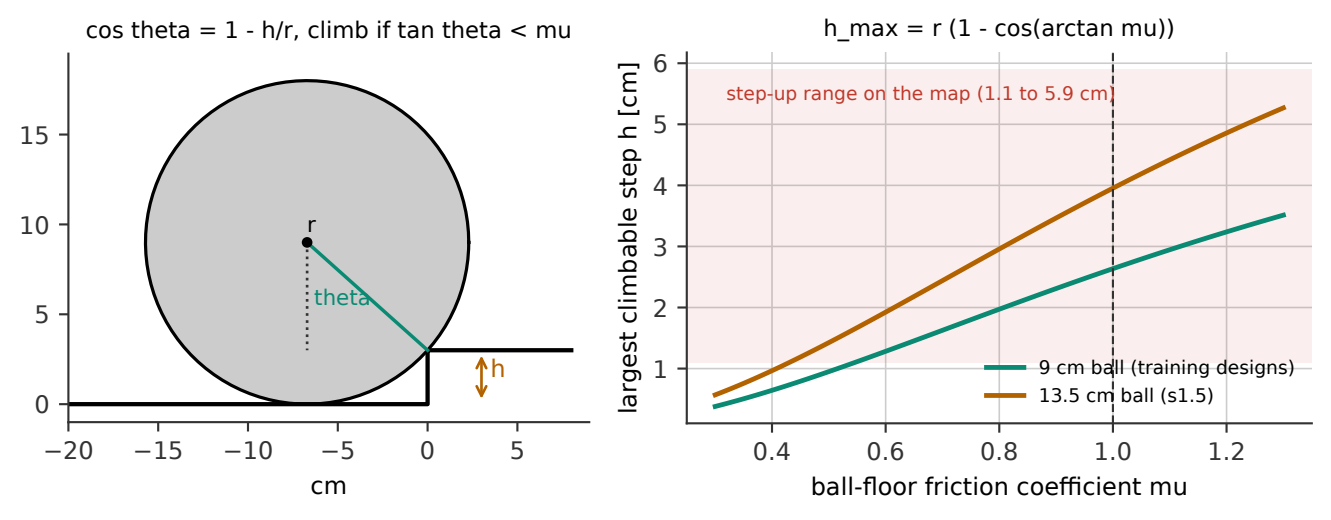
cos theta = 1 - h/r, climb if tan theta < mu
r
theta
h
15
10
5
0
−20
−15
−10
−5
0
5
cm
h_max = r (1 - cos(arctan mu))
step-up range on the map (1.1 to 5.9 cm)
9 cm ball (training designs)
13.5 cm ball (s1.5)
largest climbable step h [cm]
6
5
4
3
2
1
0.4
0.6
0.8
1.0
1.2
ball-floor friction coefficient mu